Course Code	Course Title	L	T	P	J	C
CSE1907	Comprehensive Examination	0	0	0	0	1
Pre-requisite	NIL	Syllabus version				
		1.0				

Course Objectives:

To evaluate the overall understanding of the students in the core areas of M. Tech CSE and Business Analytics

Course Outcomes:

At the end of the course, the student will be able to

1. Define, explain, evaluate, and interpret the fundamental knowledge pertaining to the field domain of Computer science and Engineering and apply that essential knowledge to the field of Business Analytics.

Digital Logic and Computer Organization

Simplification of Boolean functions– Combinational logic: Adder, subtractor, encoder, decoder, multiplexer, de-multiplexer – Sequential Logic: Flip flops- Instructions - Instruction types- Instruction Formats - Addressing Modes- Pipelining- Data Representation - Memory Hierarchy- Cache memory-Virtual Memory- I/O Fundamentals- I/O, Techniques - Direct Memory Access - Interrupts-RAID architecture

Programming, Data Structures and Algorithms

Programming in C; Algorithm Analysis — Iterative and Recursive Algorithms; ADT - Stack and its Applications - Queue and its Applications; Data Structures — Arrays and Linked Lis'ts; Algorithms - Sorting — Searching; Trees — BST, AVL; Graphs — BFS , DFS , Dijkstra's Shortest Path Algorithm.

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Operating Systems

Processes, Threads, Inter-process communication, CPU scheduling, Concurrency and synchronization, Deadlocks, Memory management and Virtual memory & File systems.

Database Management System

DBMS, Schema, catalog, metadata, data independence, pre-compiler; Users-naïve, sophisticated, casual ;ER Model- Entity, attributes, structural constraints; Relational Model-Constraints, Relational Algebra operations; SQL- DDL, DML, TCL, DCL commands, basic queries, Normalization-properties, 1NF, 2NF, 3NF, BCNF; Indexing-different types, Hash Vs B-tree Index; Transaction-problems, Concurrency Control-techniques, Recovery-methods.

Computer network and Cryptography

Circuit Switching, Packet Switching, Frame Relay, Cell Switching, ATM , OSI Reference model, TCP\IP, Network topologies, LAN Technologies, Error detection and correction techniques, Internet protocols , IPv4/IPv6, Routing algorithms, TCP and UDP, Sockets, Congestion control, Application Layer Protocols, Network Security: Basics of public and private key cryptosystems- Digital Signatures and Key Management, User Authentication, Kerberos, Firewalls.
Machine and Deep Learning
Linear Regression, Multiple Linear Regression,Logistic Regression, Linear and Non Linear, Kernel Functions, K- Nearest Neighbors, K-means Clustering, Self-Organizing Map, Random Forest Trees, Naïve Bayes Classifier, Semi-Supervised Learning, Unsupervised Training of Neural Networks, CNN Architectures, RCNN (Region Based CNN),YOLO(You Only Look Once), Dense Net, Deep-Q-Nets
Data Science and Business Analytics
Data Science Methodology, Competing on Analytics with Internal and external Processes, Data Visualization and Analytics, Linear Regression, Linear optimization, Linear algebra with Numpy. Pandas series Object, Pandas data Frame, Reading and writing the data including JSON data, Continuous and Discrete Probability distribution, Exploratory data analysis.
Big Data computing and business analytics
Big Data Analytics-Typical Analytical Architecture, Hadoop Frame work, Spark Context, Resilient Distributed Datasets, Creating RDD, Data Analytics Life Cycle–Main phases of the lifecycle, Association Rules – Sequence Rules, Batch Analysis –with Hadoop Map Reduce, Graph Analytics Techniques, Social Network Metrics

Recommended by Board of Studies	13-06-2023		
Approved by Academic Council	No. 70	Date	24-06-2023